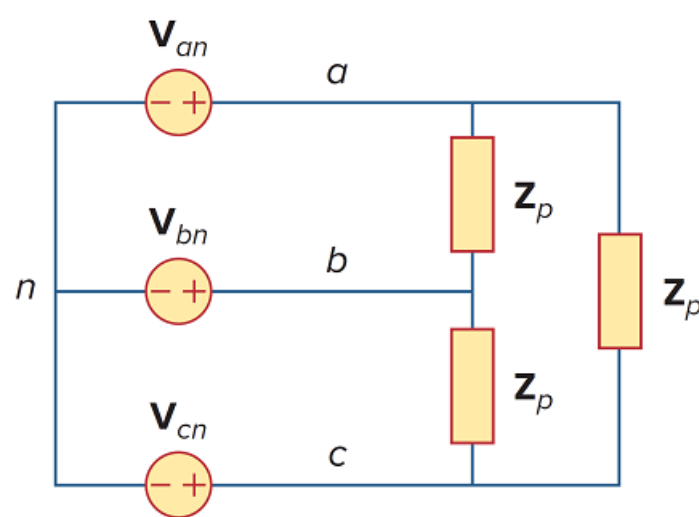


Problem

In the Y-Δ system shown in Fig. 12.44, the source is a positive sequence with

$\mathbf{V}_{an} = 440\angle 0^\circ \text{ V}$ and phase impedance $\mathbf{Z}_p = 2 - j3 \Omega$. Calculate the line voltage $\mathbf{V}_L$ and the line current $\mathbf{I}_L$.

Figure 12.44



Step-by-step solution

Step 1 of 2

Refer to Figure 12.44 in the textbook.

The phase voltage is, $\mathbf{V}_{an} = 440\angle 0^\circ \text{ V}$.

Calculate the line voltage.

$$\begin{aligned}\mathbf{V}_L &= \sqrt{3}\mathbf{V}_{an}\angle 30^\circ \\ &= (\sqrt{3})440\angle (0^\circ + 30^\circ) \\ &= 762.1\angle 30^\circ \text{ V}\end{aligned}$$

Hence, the line voltage is $\boxed{762.1\angle 30^\circ \text{ V}}$.

Step 2 of 2

The phase impedance is, $Z_p = 2 - j3 \Omega$.

When all the delta connected load have same value, convert the delta load into star load as follows:

$$\begin{aligned} Z_{star} &= \frac{Z_{\Delta}}{3} \\ &= \frac{Z_p}{3} \\ &= \frac{2 - j3}{3} \\ &= 1.202 \angle -56.30^\circ \Omega \end{aligned}$$

Calculate the line current.

$$\begin{aligned} I_L &= \frac{V_{an}}{Z_{star}} \\ &= \frac{440 \angle 0^\circ}{1.202 \angle -56.30^\circ} \\ &= 366.1 \angle 56.30^\circ \text{ A} \end{aligned}$$

Hence, the line current is $\boxed{366.1 \angle 56.30^\circ \text{ A}}$.